

c) If $f(x) = x^4 - 8x^3$

a) Classify all local extrema using the Second Derivative Test. If the Second Derivative Test fails, use the First Derivative Test.

b) Determine the Points of Inflection by creating an interval chart for $f''(x)$

a) $f'(x) = 4x^3 - 24x^2$ Interval $(-\infty, 0)$ $(0, 6)$ $(6, \infty)$

$= 4x^2(x-6)$

critical #'s: $x=0, 6$

$f''(x) = 12x^2 - 48x$

$f''(0) = 0$ test fails

$f''(6) = 144 > 0$

$\Rightarrow$ min. @ $(6, -432)$

x^2	+	+	
$x-6$	-	-	
$f'(x)$	-	-	
$f(x)$	$\downarrow$	$\downarrow$	

$\therefore$ not a local extrema at $x=0$.

b) Interval $(-\infty, 0)$ $(0, 4)$ $(4, \infty)$

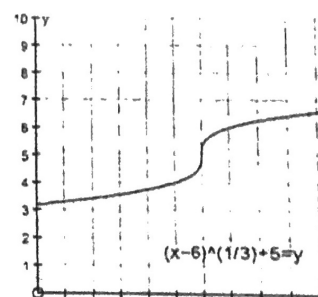
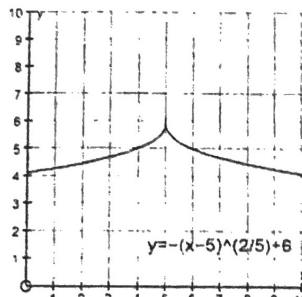
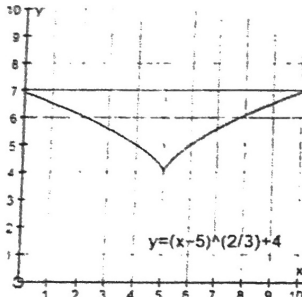
x	-	+	+
$x-4$	-	-	+
$f''(x)$	+	-	+
$f(x)$	c.u.	c.d.	c.u.

$\therefore$ poi @ $(0, 0)$ & $(4, -256)$

A word on Cusps

$f''(x) = 12x(x-4)$
 $x=0, 4$

A cusp is a pointy bend on a graph where no tangent line exists. A cusp is a good example of the situation where a function can have a maximum or minimum at a point where the derivative of the function does not exist.



To spot a cusp

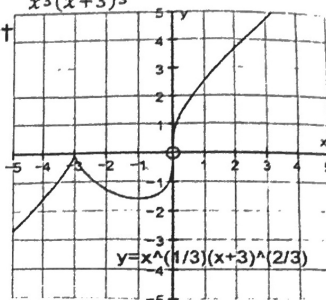
- 1) Determine where the derivative does not exist and test the concavity on either side of that critical number.
- 2) If the concavity remains the same, you have found a cusp (first 2 pictures). If the concavity changes, you have found a P.O.I with a vertical tangent (3rd picture).

Ex) Verify that there is a cusp on the graph of the function $f(x) = x^{2/3}$ when $x = 0$.

This question was done in class!

Ex) For the function $f(x) = x^{\frac{1}{3}}(x+3)^{\frac{2}{3}}$, $f'(x) = \frac{x+1}{x^{\frac{2}{3}}(x+3)^{\frac{1}{3}}}$, and $f''(x) = -\frac{5}{x^{\frac{5}{3}}(x+3)^{\frac{4}{3}}}$ critical #'s: $-3, -1, 0$.

- Find the intervals of increase and decrease using an interval chart
- Find the local maximum and minimum points
- Find the intervals of concavity using an interval chart
- Find the points of inflection
- Label the points you found in b) and d) on the graph to the right



Interval: $(-\infty, -3)$, $(-3, -1)$, $(-1, 0)$, $(0, \infty)$

$x+1$	-	-	+	+
$x^{\frac{2}{3}}$	+	+	+	+
$(x+3)^{\frac{1}{3}}$	-	+	+	+
$f'(x)$	+	-	+	+
$f(x)$	↑	↓	↑	↑

Interval: $(-\infty, -3)$, $(-3, 0)$, $(0, \infty)$

-2	-	-	-
$x^{\frac{5}{3}}$	-	-	+
$(x+3)^{\frac{4}{3}}$	+	+	+
$f''(x)$	+	+	-
$f(x)$	c.u.	c.u.	c.d.

$\therefore$ Max @ $(-3, 0)$; min @ $(-1, -\sqrt[3]{4})$

Increasing Decreasing

$(-\infty, -3) \cup (-1, \infty)$ $(-3, -1)$ $\therefore$ poi @ $(0, 0)$

cusp @ $(-3, 0)$

c.u. @ $(-\infty, 0)$

c.d. @ $(0, \infty)$

Ex) For the curve $f(x) = 2 - 12x + 9x^2 - 2x^3$.

- Find the local maximum and minimum points using an interval chart
- Find any points of inflection using an interval chart

$$a) f'(x) = -6x^2 + 18x - 12$$

$$= -6(x-2)(x-1)$$

critical #'s at $x=2, 1$.

Interval: $(-\infty, 1)$, $(1, 2)$, $(2, \infty)$

-6	-	-	-
$x-2$	-	-	+
$x-1$	-	+	+
$f'(x)$	-	+	-
$f(x)$	↓	↑	↓

Local Max @ $(2, -2)$

Local Min @ $(1, -3)$

$$b) f''(x) = -12x + 18$$

$$0 = -12(x - \frac{3}{2})$$

$$\therefore x = \frac{3}{2}$$

Interval: $(-\infty, \frac{3}{2})$, $(\frac{3}{2}, \infty)$

$-12x + 18$	+	-
$f''(x)$	+	-
$f(x)$	c.u.	c.d.

$\therefore$ poi @ $(\frac{3}{2}, -\frac{5}{2})$